MIL-I-60531 (MU)
15 August 1967

SUPERSEDING
FA-PD-MI-2588, Rev. O
29 March 1963

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, M89
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, M89 Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel.

DRAWINGS

Munitions Command

D10522696	- Initiator, Cartridge Actuated, Delay, M89 Metal Parts Assembly
C8846785	- Container, Ammunition, Fiber M459 for Initiator, Cartridge Actuated Delay

FSC 1377

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Drawings (Cont'd)

C8846786 - Box, Packing, Ammunition for Initiators, Cartridge Actuated, Delay in Fiber Container M459.

IEL 10522696 - Inspection Equipment List for Initiator Cartridge Actuated, Delay, M89, Metal Parts Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified in the applicable drawings, referenced specifications and the following:

3.2 First article sample.- Unless otherwise directed by the contracting officer, first article sample shall be required.

3.3 Mechanical properties.- Mechanical properties required by the drawings shall be uniform through out the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure.- The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.1).

3.5 Firing pin release.- The firing pin of the metal parts assembly, less safety pin shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inches, minimum, and .425 inches, maximum.

3.6 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.7 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component, regardless of configuration.

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3.8 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.9 Workmanship.- The requirements for workmanship are as specified on the applicable drawing, referenced specifications and the following:

3.9.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.9.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.9.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 50 units (unplated) of each heat treated component for hardness tests, 5 assemblies and 10 sets of unassembled component for visual and dimensional examination and shall be tested as indicated in Table I.

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Table I

Paragraphs Requirements	Tests	Type of Test	Sample Size	Dest	Non-Dest
3.3	4.4.1	Hardness	50	X	-
3.4	4.4.2	Hydrostatic pressure	each	-	X
3.5	4.4.3	Firing pin release	each	-	X
3.6	4.4.4	Initiator pin security	each	-	X
3.7	4.4.5	Protective finish as specified in reference specifications		X	-
3.8	4.4.6	Residual magnetism	each	-	X

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility as specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirement of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot of metal parts assemblies shall consist of 1200 initiators, maximum, provided the homogenity requirements of inspection lots have been retained.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables II and III. For an individual defect (single characteristic) the criteria in Table II shall be used. For total defects with a major or minor group the

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4.3.2 (Cont'd)

criteria in Table III shall be used. All non-conforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table II

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

Table III

	Total Allowable Defects per Sample Size			
Sample Size for Major Defects Number of defects in class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 16	3	4	5	7
17 - 48	4	5	8	10
over 48	6	8	11	14
Sample Size for Minor Defects Number of defects in class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Chamber, Initiator, Drawing C8597728, covering a detail of Drawing D10522696.

Defect	Method of Inspection
CRITICAL	
None defined	

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4.3.2.1 (Cont'd)

	Defect	Method of Inspection
MAJOR		
101.	Pitch diameter of large external thread, min.	Gage
102.	Major diameter of large external thread, min.	Gage
103.	Pitch diameter of small external thread, min.	Gage
104.	Major diameter of small external thread, min.	Gage
105.	Pitch diameter of internal thread, max.	Gage
106.	Minor diameter of internal thread, max.	Gage
107.	Concentricity of datum location with pitch diameter of small external thread	Gage
108.	Concentricity of pitch diameter of large external thread with cavity	Gage
109.	Concentricity of large external thread pilot with large external thread	Gage
110.	Concentricity of counterbore with cavity including perpendicularity with datum location at mouth	Gage
111.	Diameter of counterbore	Gage
112.	Diameter of large thread pilot	Gage
113.	Depth of cavity	Gage
114.	Length to datum location on gas port	Gage
115.	Thickness of wall, min.	Gage
116.	Metal defective	Visual
117.	Protective finish inadequate or defective	Visual
118.	Surface finish of cavity and finish on angle of gas port incorrect	Visual
119.	Burr or sharp corner	Visual
120.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of large external thread, max.	Gage
202.	Pitch diameter of small external thread, max.	Gage
203.	Pitch diameter of internal thread, min.	Gage
204.	Diameter of small external thread relief	Gage
205.	Diameter of small external thread pilot	Gage
206.	Diameter of hole in gas port	Gage
207.	Diameter of cavity	Gage
208.	Diameter of chamfer at internal thread	Gage
209.	Depth of threaded hole	Gage
210.	Total length	Gage
211.	Length of large external thread	Gage
212.	Length of gas port	Gage
213.	Length of small thread relief	Gage

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
214. Length from face of gas port to small thread	Gage
215. Length of large external thread pilot	Gage
216. Thickness of wall, max.	Gage
217. Location of datum diameter, at mouth	Gage
218. Radius or chamfer missing or incorrect	Visual
219. Surface finish improper, except as otherwise classified	Visual

4.3.2.2 Pin, Firing, Drawing C10520411, covering a detail of Drawing D10522696.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of large cavity with body	Gage
102. Concentricity of striker with body	Gage
103. Concentricity of small cavity with body	Gage
104. Concentricity of ball hole counterbore with ball holes	Gage
105. Diameter of striker	Gage
106. Diameter of body	Gage
107. Diameter of ball hole, max.	Gage
108. Diameter of ball hole counterbore	Gage
109. Diameter of pin relief holes, min.	Gage
110. Depth of ball counterbore, min.	Gage
111. Length of striker, min.	Gage
112. Location of ball hole counterbore	Gage
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish of body improper	Visual
116. Burr or sharp corner	Visual
117. Foreign matter	Visual
MINOR	
201. True position of relief holes with small cavity	Gage
202. Diameter of large cavity	Gage
203. Diameter of small cavity	Gage
204. Diameter of ball hole, min.	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MINOR (cont'd)	
205. Diameter of pin relief holes, max.	Gage
206. Depth of large cavity	Gage
207. Depth to bottom of small cavity	Gage
208. Depth of ball hole counterbore, max.	Gage
209. Length of striker, max.	Gage
210. Total length	Gage
211. Radius or chamfer missing or incorrect	Visual
212. Surface finish improper except as otherwise classified	Visual

4.3.2.3 Initiator Spring, Drawing B10520412, covering a detail of Drawing D10522696.

CRITICAL

None defined

MAJOR

101. Load at specified heights	Gage
102. Closed length, max.	Gage
103. Ends incorrect	Visual
104. Protective finish inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual

MINOR

None defined

4.3.2.4 Filter, Drawing C10520414, covering a detail of Drawing D10522696.

CRITICAL

None defined

MAJOR

101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with head and center hole	Gage

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
104. True position of vent holes with diameter of head	Gage
105. Diameter of center hole	Gage
106. Diameter of spaced holes	Gage
107. Diameter across threads, max.	Gage
108. Depth of slot in head, max.	Gage
109. Depth of center hole	Gage
110. Thickness of wall at base of self lock element slot	Gage
111. Number of holes incorrect	Visual
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Burr	Visual
115. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of head	Gage
203. Diameter of undercut	Gage
204. Depth of slot in head, min.	Gage
205. Length of head	Gage
206. Total length	Gage
207. Width of undercut, max.	Gage
208. Width of slot in head	Gage
209. Location of slot for self locking element	Gage
210. Location of slot in head	Gage
211. Chamfer missing or incorrect	Visual
212. Surface finish, improper	Visual

4.3.2.5 Seal, Drawing B10520415, covering a detail of Drawing D10522696.

CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Width	Gage
103. Thickness, min.	Gage
104. Material defective	Visual
105. Flash	Visual
106. Foreign matter	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter, max.	Gage
202. Thickness, max.	Gage

4.3.2.6 Pin, Initiator, Drawing C10520553, covering a detail of Drawing D10522696.

CRITICAL	
None defined	
MAJOR	
101. Concentricity of groove in head with shoulder	Gage
102. Concentricity of body with shoulder	Gage
103. Concentricity of head with shoulder	Gage
104. Concentricity of width of pull hole end with shoulder	Gage
105. Symmetry of thickness of pull hole end with shoulder	Gage
106. Symmetry of pull hole with width of pull hole end	Gage
107. Diameter of undercut, min., if applicable	Gage
108. Width of undercut, max., if applicable	Gage
109. Diameter of body	Gage
110. Diameter of shoulder	Gage
111. Diameter of groove on body, min.	Gage
112. Width of head groove	Gage
113. Location of groove on head	Gage
114. Location of groove on body	Gage
115. Location of pull hole	Gage
116. Profile of end	Visual
117. Profile of head	Visual
118. Metal defective	Visual
119. Protective finish inadequate or defective	Visual
120. Surface finish of groove in head and face of shoulder improper	Visual
121. Burr	Visual
122. Foreign matter	Visual
MINOR	
201. Diameter of groove on head	Gage
202. Diameter of groove on body, max.	Gage
203. Diameter of pull hole	Gage

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
204. Length of groove on body	Gage
205. Total length	Gage
206. Width of flat at body groove end	Gage
207. Thickness of pull hole end	Gage
208. Radius or chamfer missing or incorrect	Visual
209. Surface finish improper, except as otherwise classified	Visual

4.3.2.7 Firing Pin Housing, Drawing C10522693, covering a detail of Drawing D10522696.

CRITICAL	
1. Diameter of striker hole, min.	Gage
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of striker hole with cavity	Gage
104. Concentricity of cavity with pitch diameter of thread including perpendicularity with rear of flange	Gage
105. Concentricity of body with pitch diameter of thread including perpendicularity with rear of flange	Gage
106. Concentricity of chamfer of striker hole with cavity	Gage
107. Parallelism of face with rear flange	Gage
108. Diameter of body, min.	Gage
109. Diameter of cavity	Gage
110. Diameter of flange, min.	Gage
111. Diameter of undercut, min.	Gage
112. Total length, max.	Gage
113. Width of undercut	Gage
114. Thickness of striker hole	Gage
115. Location of datum diameter	Gage
116. Location of thread	Gage
117. Chamfer at striker hole incorrect	Gage
118. Height of boss	Gage
119. Metal defective	Visual
120. Protective finish inadequate or defective	Visual
121. Surface finish of cavity improper	Visual
122. Burr or sharp corner	Visual
123. Foreign matter	Visual

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of striker hole, max.	Gage
204. Diameter of flange, max.	Gage
205. Diameter of wrench holes	Gage
206. Depth of wrench holes	Gage
207. Length of body to flange	Gage
208. Total length, min.	Gage
209. Location of wrench holes	Gage
210. Radius or chamfer missing or incorrect, except as otherwise classified	Visual
211. Surface finish improper, except as otherwise classified	Visual

4.3.2.8 Cap, Drawing F10522694, covering a detail of Drawing D10522696.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of small internal thread, max.	Gage
102. Minor diameter of small internal thread, max.	Gage
103. Pitch diameter of large internal thread, max.	Gage
104. Minor diameter of large internal thread, max.	Gage
105. Concentricity of pitch diameter of large internal thread with midsection cavity including perpendicularity with base of counterbore of large cavity	Gage
106. Concentricity of pitch diameter of small internal thread with midsection cavity including perpendicularity with base of counterbore of large cavity	Gage
107. Concentricity of hole between small cavity and boss cavity with midsection cavity including perpendicularity with base of large cavity	Gage
108. Concentricity of large cavity with pitch diameter of large internal thread	Gage
109. Concentricity of boss cavity with hole between small cavity and boss cavity	Gage
110. Concentricity of flange with hole between small cavity and boss cavity	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MAJOR (Cont'd)	
111. Concentricity of body, large end with large internal thread	Gage
112. Concentricity of small cavity with midsection cavity	Gage
113. Symmetry of thickness of mounting lugs with body, large end	Gage
114. True portion of mounting holes with small hole adjacent to boss cavity	Gage
115. Diameter of large cavity rear of thread run-out	Gage
116. Diameter of small cavity	Gage
117. Diameter of boss cavity, max.	Gage
118. Diameter of hole between small cavity and boss cavity, min.	Gage
119. Diameter of safety pin hole, max.	Gage
120. Diameter of flange	Gage
121. Depth from base of large cavity to face of internal boss	Gage
122. Length of effective small internal thread, min.	Gage
123. Length of small internal thread, max.	Gage
124. Length of effective large internal thread, min.	Gage
125. Length of internal large thread, max.	Gage
126. Thickness of wall, min.	Gage
127. Location of safety pin holes	Gage
128. Height of boss	Gage
129. Radial area of lugs not continuous or not tangent to body	Visual
130. Metal defective	Visual
131. Protective finish inadequate or defective	Visual
132. Burr or sharp corner	Visual
133. Foreign matter	Visual
MINOR	
201. Pitch diameter of set screw thread	Gage
202. Minor diameter of set screw thread	Gage
203. Pitch diameter of small internal thread, min.	Gage
204. Pitch diameter of large internal thread, min.	Gage
205. Diameter of body, small end	Gage
206. Diameter of body, large end	Gage
207. Diameter of groove, small end	Gage
208. Diameter of chamfer, small internal thread	Gage

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4.3.2.8 (Cont'd)

	Defect	Method of Inspection
MINOR (Cont'd)		
209.	Diameter of midsection cavity	Gage
210.	Diameter of boss cavity, min.	Gage
211.	Diameter of hole between boss cavity and small cavity, max.	Gage
212.	Diameter of safety pin hole, min.	Gage
213.	Diameter of mounting hole	Gage
214.	Depth from base of large cavity to base of small cavity	Gage
215.	Depth to bottom of large cavity rear to runout	Gage
216.	Depth of large cavity	Gage
217.	Total length	Gage
218.	Length of body, large end	Gage
219.	Width of groove, small end	Gage
220.	Width across flat	Gage
221.	Width of headspace at internal boss	Gage
222.	Thickness of mounting flange	Gage
223.	Location of set screw hole	Gage
224.	Location of mounting holes	Gage
225.	Location of lugs	Gage
226.	Location of flats	Gage
227.	Length from rear flange to shoulder	Gage
228.	Width of flange	Gage
229.	Safety pin hole obstructed	Visual
230.	Radius or chamfer missing or incorrect	Visual
231.	Surface finish improper	Visual

4.3.2.9 Safety Pin, Drawing B10522817, covering a detail of Drawing D10522696.

CRITICAL

None defined

MAJOR

101.	Diameter of wire	Gage
102.	Profile incorrect	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

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4.3.2.9 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of head, outside	Gage
202. Length between high and low ends	Gage
203. Total length	Gage
204. Location of center loop	Gage
205. Radius or chamfer of center loop incorrect	Visual

4.3.2.10 Assembly, Drawing D10522696, less all drawings except C10520411, C10522693 and MS19060-3.

CRITICAL	
1. Firing pin protrusion	Gage
2. Steel ball missing	Gage
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.11 Assembly, Drawing D10522696.

CRITICAL	
None defined	
MAJOR	
101. Firing pin not cocked	Manual
102. Initiator pin not free to rotate	Manual
103. Any part missing except as otherwise classified	Visual
104. Improper assembly	Visual
105. Safety pin improperly inserted	Visual
106. Marking missing, incorrect or illegible	Visual
MINOR	
201. Shipping cap missing	Visual
202. Protective finish inadequate or defective	Visual

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4.3.2.12 Fiber Container, Drawing C8846785.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.13 Box, Packing, Drawing C8846786.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with 4.4.

4.3.3.1 Hardness.- Fifty units from each heat-treated lot of components shall be tested for compliance with the requirements specified on the applicable drawing. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

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4.3.3.2 Hydrostatic pressure.- Each chamber and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any subassembly to comply with the requirements of 3.4 shall be cause for rejection of the subassembly.

4.3.3.3 Firing pin release.- The initiator pin of each partial assembly, consisting of all parts except the safety pin and chamber, shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirement of 3.5 shall be cause for rejection of the subassembly.

4.3.3.4 Initiator pin security.- The initiator pin of each partial assembly, consisting of all parts except the chamber, shall be subjected to the initiator pin security test. Failure of any partial assembly to comply with the requirements of 3.6 shall be cause for rejection of the partial assembly.

4.3.3.5 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.6 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.8 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10522696.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.4.2 Hydrostatic pressure.- Prior to the application of protective finish, the chamber and cap subassembly shall be mounted in the device, using the fixtures for properly sealing all openings, and subjected to the hydrostatic pressure to determine compliance with 3.4.

4.4.3 Firing pin release.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

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4.4.4 Initiator pin security.- The subassembly shall be secured in the fixture, with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.4.5 Protective finish.- Test procedures for protective finish shall be in accordance with the military standard referenced on the drawing.

4.4.6 Residual magnetism.- The assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the assembly six inches from and at the same height as the compass (cap end at south) with the compass set in an east-west heading. The assembly shall be moved slowly past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8846785 and C8846786.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A057

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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